Hospital use among adults prescribed antidepressants in NHS Tayside

A Fleming,[1] Dr. L. Ciaccio ,[1,2] Dr. C. Marwick, [1,3] Dr. K. Barnett [1,2]

[1] University of Dundee, [2] Department of Public Health, NHS Tayside, [3]University of St. Andrew's

Background

- People prescribed antidepressants represent a large, diverse and clinically heterogeneous population[1]
- Patterns of hospital use in this population remain poorly characterised

Aim | Characterise hospital use among adults prescribed antidepressants by age, sex and socioeconomic deprivation and compare rates with the Scottish population

Methods

Retrospective observational study using anonymised linked data

1 Define the population | Jan 2019-Dec 2023

NHS Tayside residents aged ≥16 with ≥1 antidepressant prescription in the corresponding month

Eligibility reassessed monthly

2 Link prescribing and hospital records

- General acute admissions
- A&E attendances
- Psychiatric admissions

3 Calculate monthly hospital use rates

Calculate separately for each hospital setting

$$\frac{\text{Hospital contacts among adults prescribed antidepressants that month}}{\text{Number of adults prescribed antidepressants that month}} \times 1{,}000 = \text{Monthly hospital use rate}$$

Contacts per 1,000 adults prescribed antidepressants

Stratified by:

Age | 7 groups | **Sex** | Male / Female | **Deprivation** | SIMD quintiles

Compare mean monthly rates:

Sex: paired t-tests

Age / deprivation: one-way ANOVA + Tukey

p <0.05

4 Compare with Scottish population rates

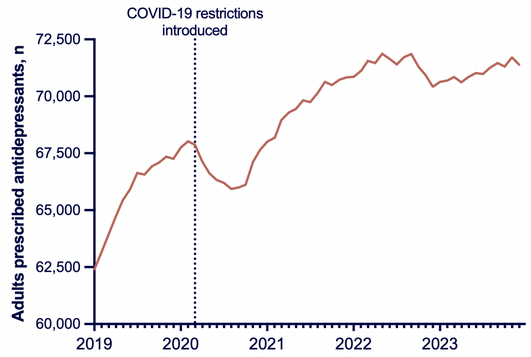
Compare age-specific and sex-specific population rates

$$\frac{\text{Tayside antidepressant-prescribed population rate}}{\text{Corresponding Scottish population rate}} = \text{Rate ratio}$$

Results

A A large and growing population are prescribed antidepressants

Mean monthly population: 68,876 adults prescribed antidepressants (1 in 5 Tayside adults), 66.1% female

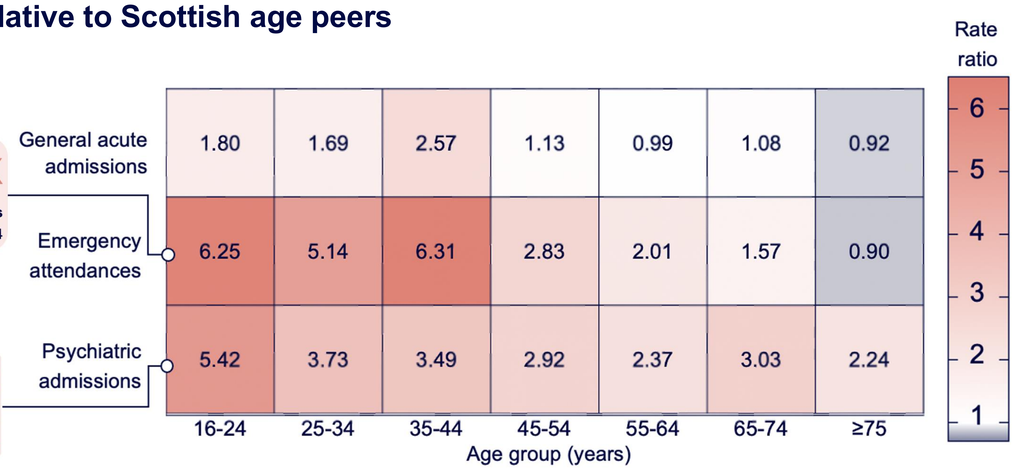


Prescribed population increased by 8.3% (Jan 2019-Dec 2023)

B Young adults prescribed antidepressants had the greatest excess hospital use relative to Scottish age peers

6.25x Emergency attendances Ages 16-24

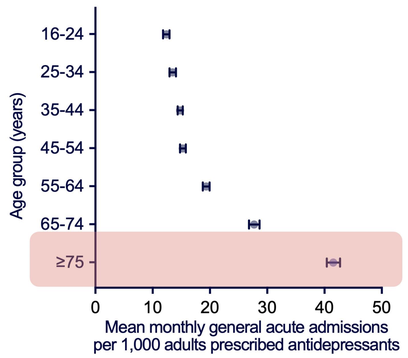
5.42x Psychiatric admissions Ages 16-24



Mean rate ratios, 2019-2023: Tayside adults prescribed antidepressants vs age-matched Scottish population

Rate ratio = 1: equal rates | >1: higher use | <1: lower use

C Highest general acute rates at ≥75: Near parity with Scottish age peers



AGE ≥75:

Absolute rate

41.86

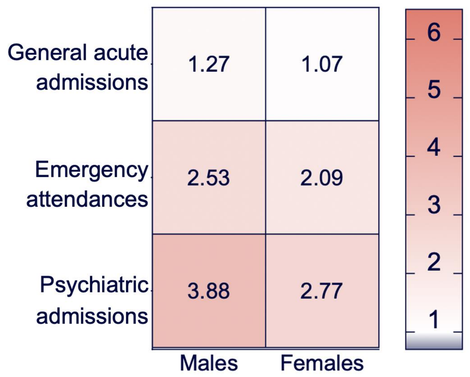
Admissions per 1,000 adults prescribed antidepressants/month

Relative rate

0.92x

The Scottish general acute admission rate for adults ≥75

D Men had greater excess hospital use

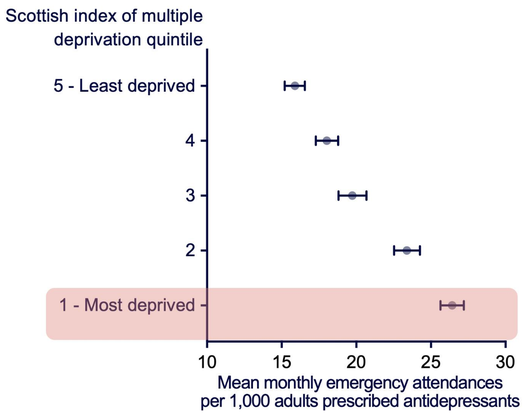


Mean rate ratios, 2019-2023: Tayside adults prescribed antidepressants Scottish population rates for the same sex.

1 = equal rates, >1 = higher use

Across all three settings, men prescribed antidepressants had greater relative excess than women, and higher absolute rates of hospital use

E Emergency attendance rates were higher with greater deprivation



66% higher

Emergency attendance rate

Most vs least deprived adults prescribed antidepressants

Discussion and Conclusions

Younger adults prescribed antidepressants had the greatest relative excess in hospital use

Those aged ≥75 had the highest general acute and emergency admission rates, but little excess

Next steps:

- Invesigate drivers of excess hospital use among adults prescribed antidepressants, particularly young adults, alongside the deprivation gradient
- Use age, sex and deprivation patterns to guide service planning for adults prescribed antidepressants

Explore this study

Reference:

1. Brisnik V, Rottenkolber M, Vukas J, Schechner M, Lukaschek K, Jung-Sievers C, et al. Potential deprescribing indications for antidepressants between 2012 and 2019: repeated cross-sectional analysis in two Scottish health boards. BMC Med. 2024;22:378. doi:10.1186/s12916-024-03584-9